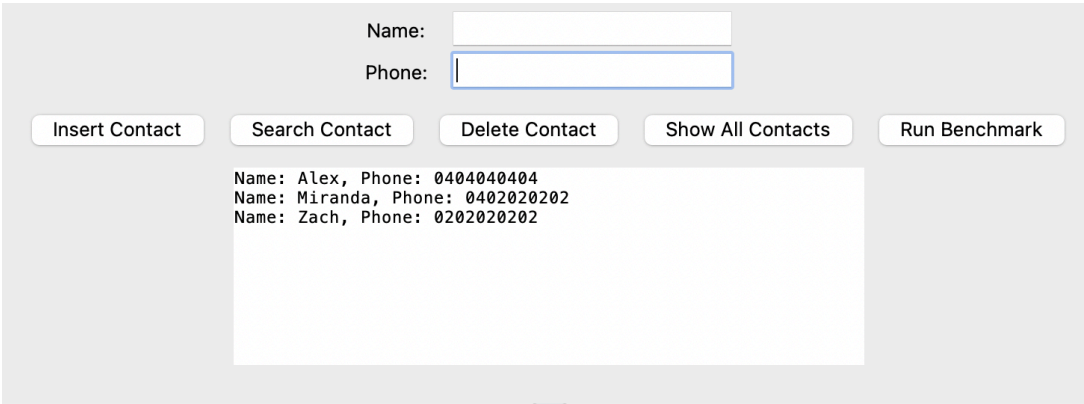
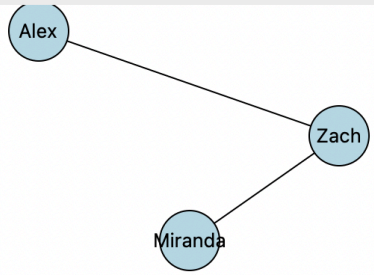
## Task 1: Output

```
All contacts:
Name: Alice, Phone: 123-456-7890
Name: Bob, Phone: 234-567-8901
Name: Eve, Phone: 345-678-9012

Searching for Bob:
Found contact: Bob - 234-567-8901

Deleting Alice:
Name: Bob, Phone: 234-567-8901
Name: Eve, Phone: 345-678-9012

Process finished with exit code 0
```

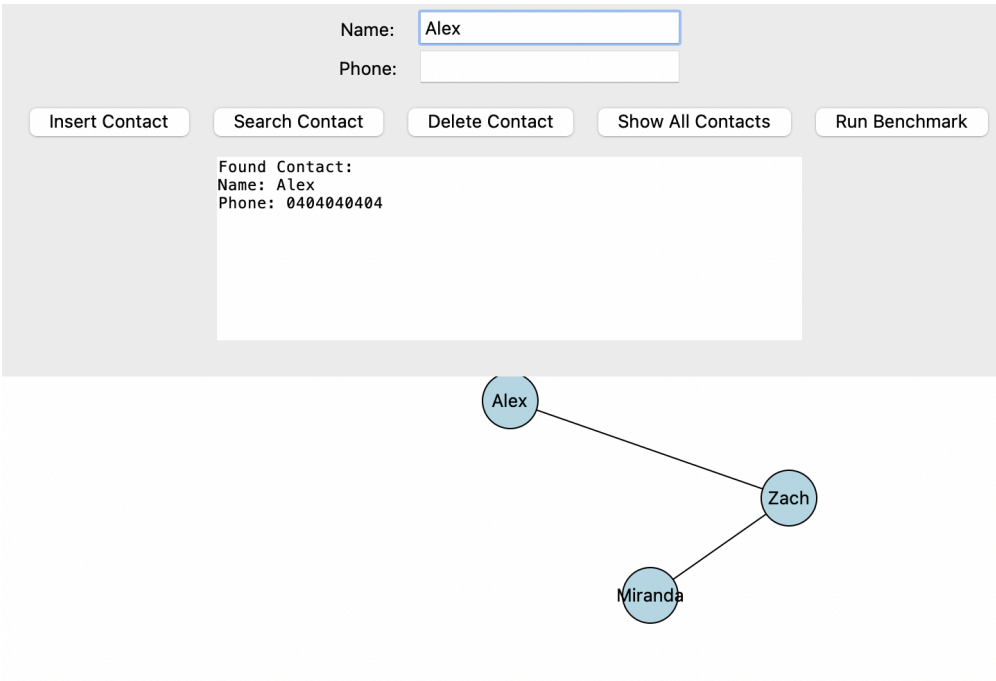
## Task 2:

### Insert

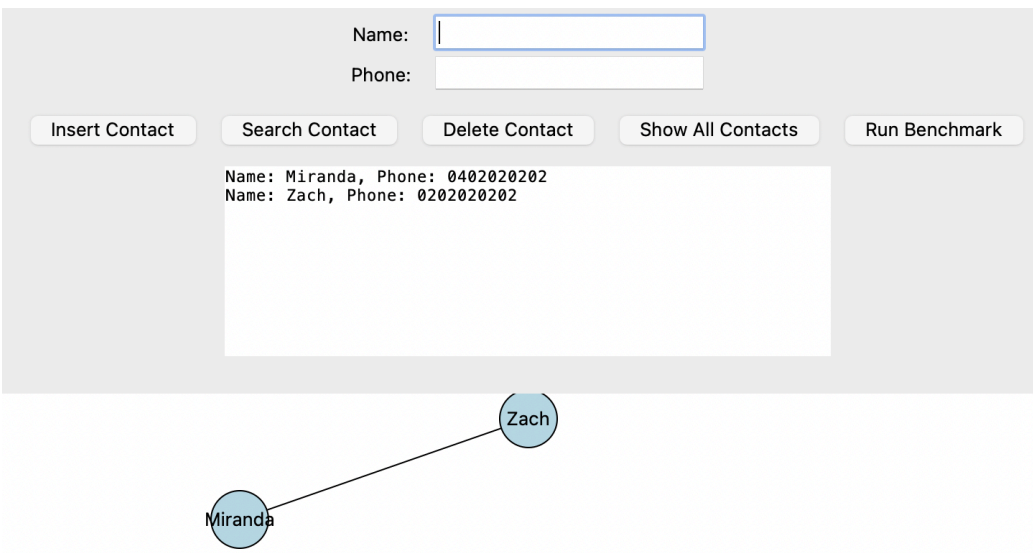


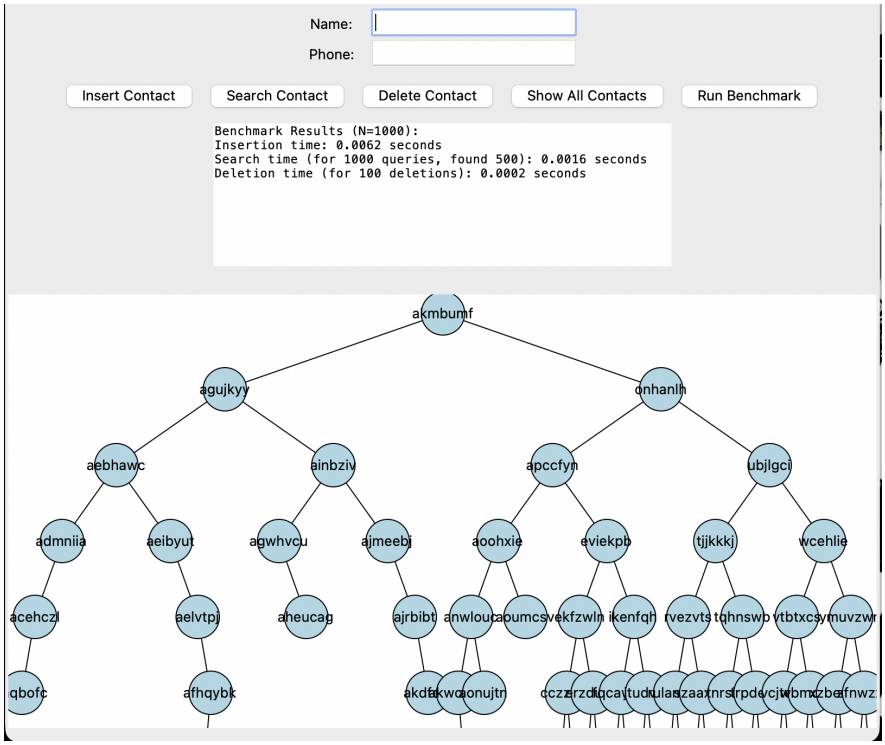
## Search



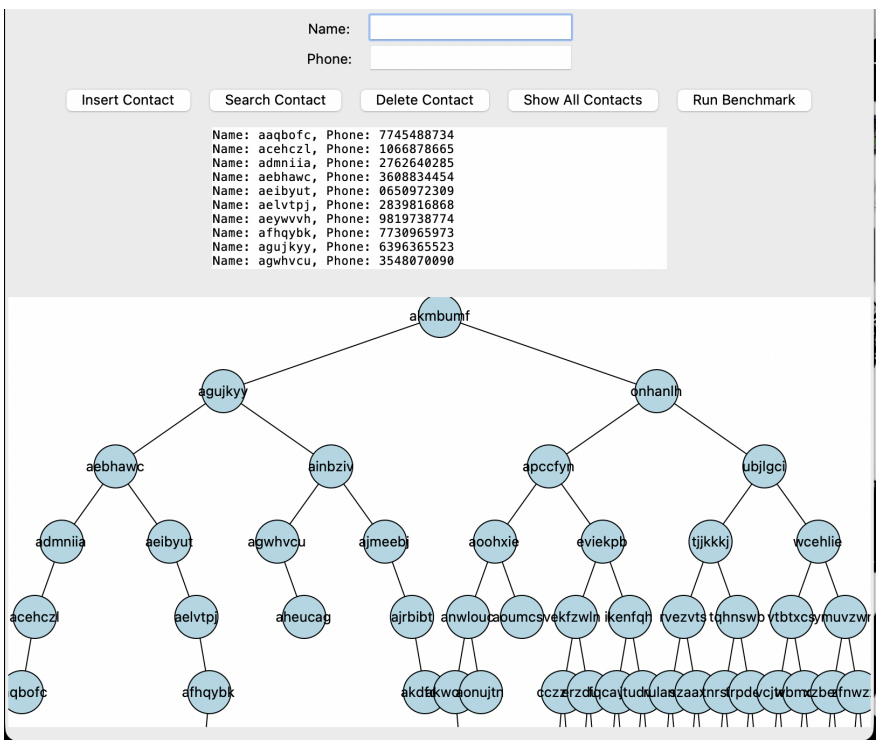

## Delete



## Benchmark

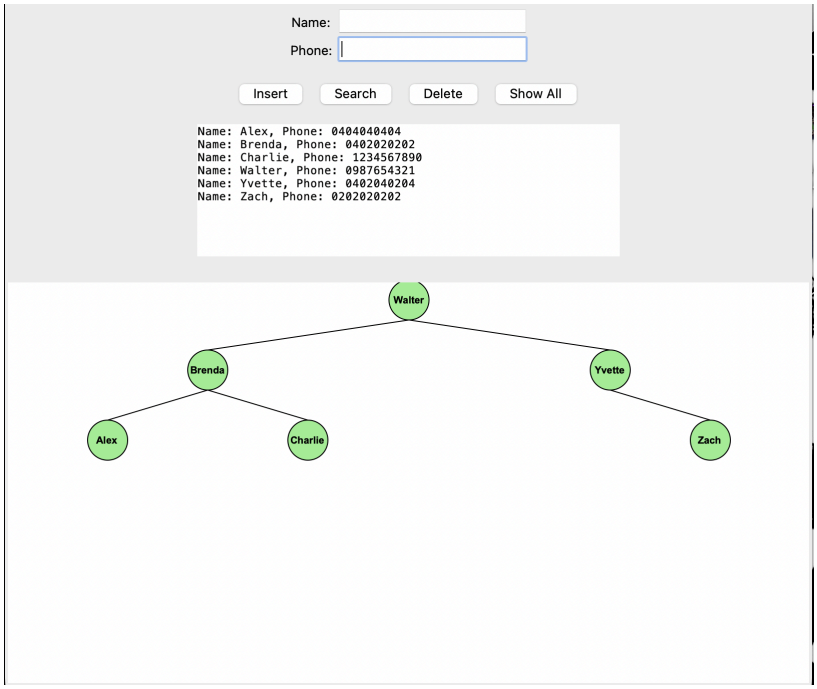



## Show All Contacts

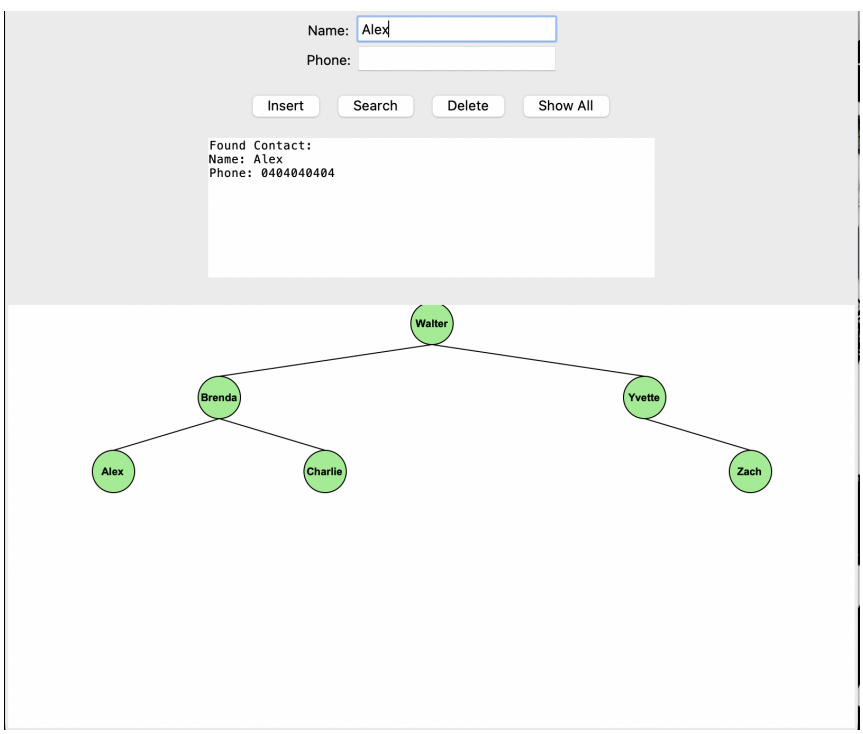


# Task 3:

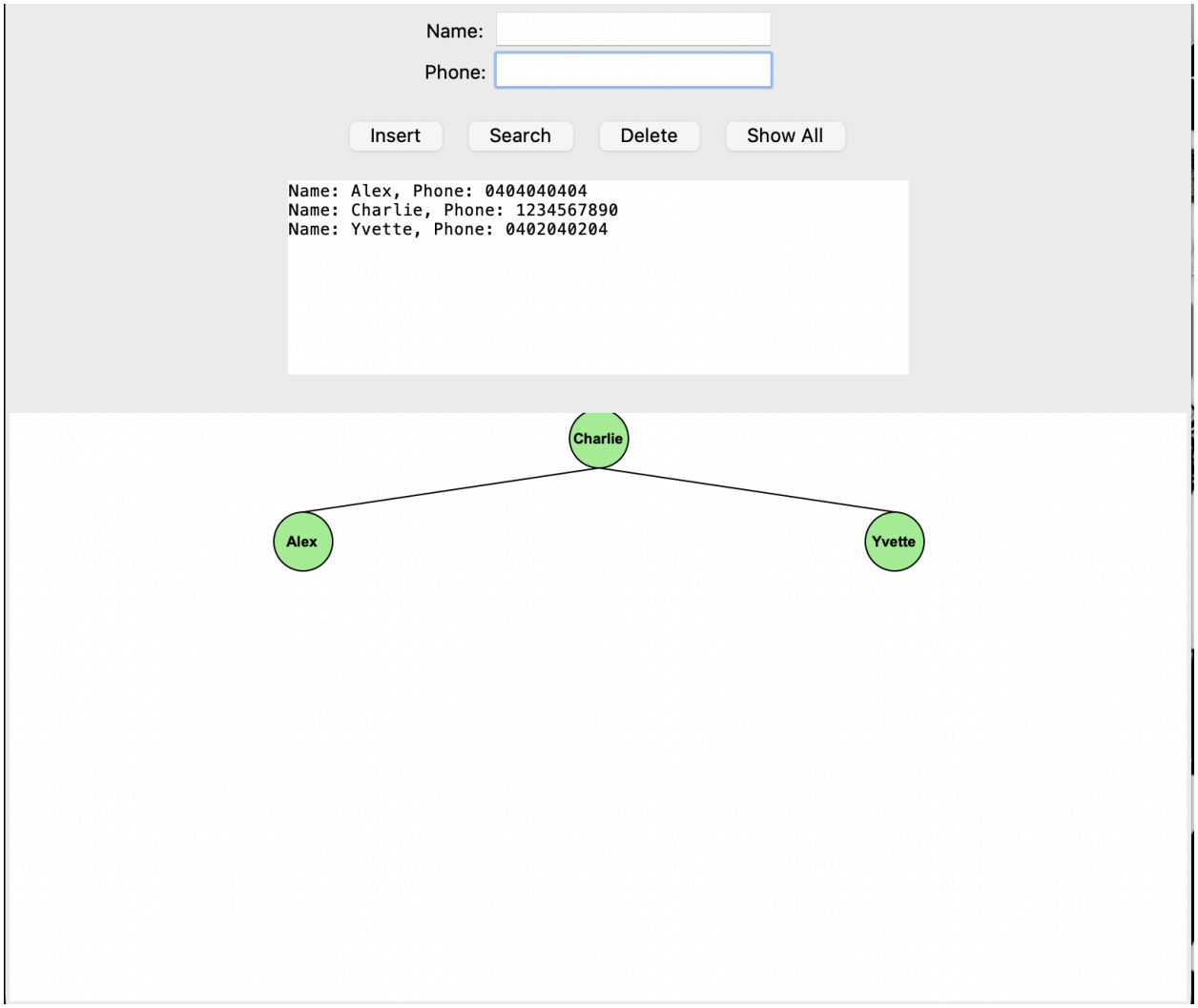
## Insert




## Search



## Delete

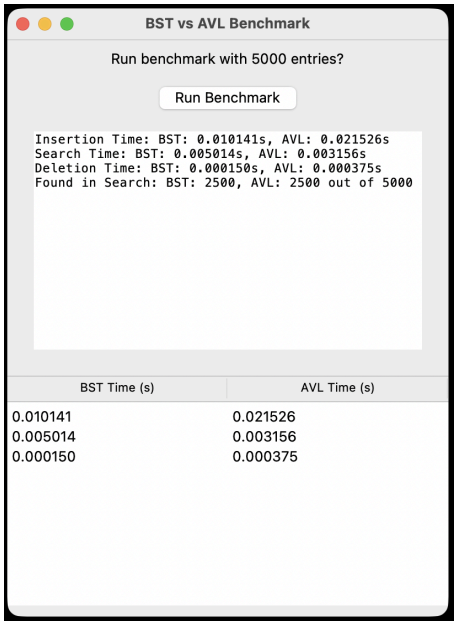

Name:
Phone:
Insert
Search
Delete
Show All
Name: Alex, Phone: 0404040404
Name: Charlie, Phone: 1234567890
Name: Yvette, Phone: 0402040204
Charlie
Alex
Yvette

# Task 4:

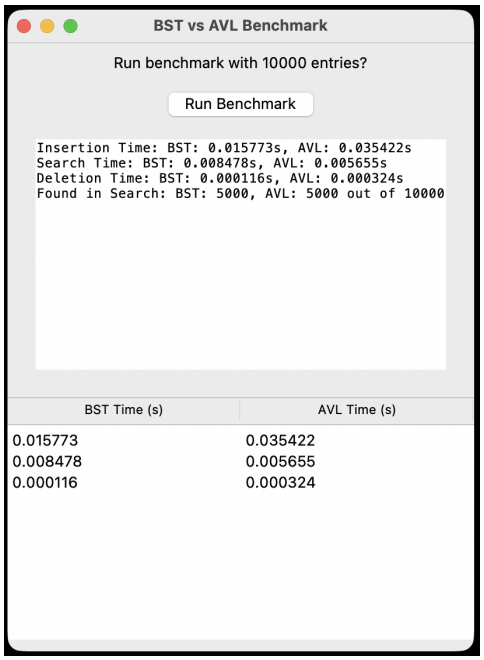
## Benchmark App

### 1000

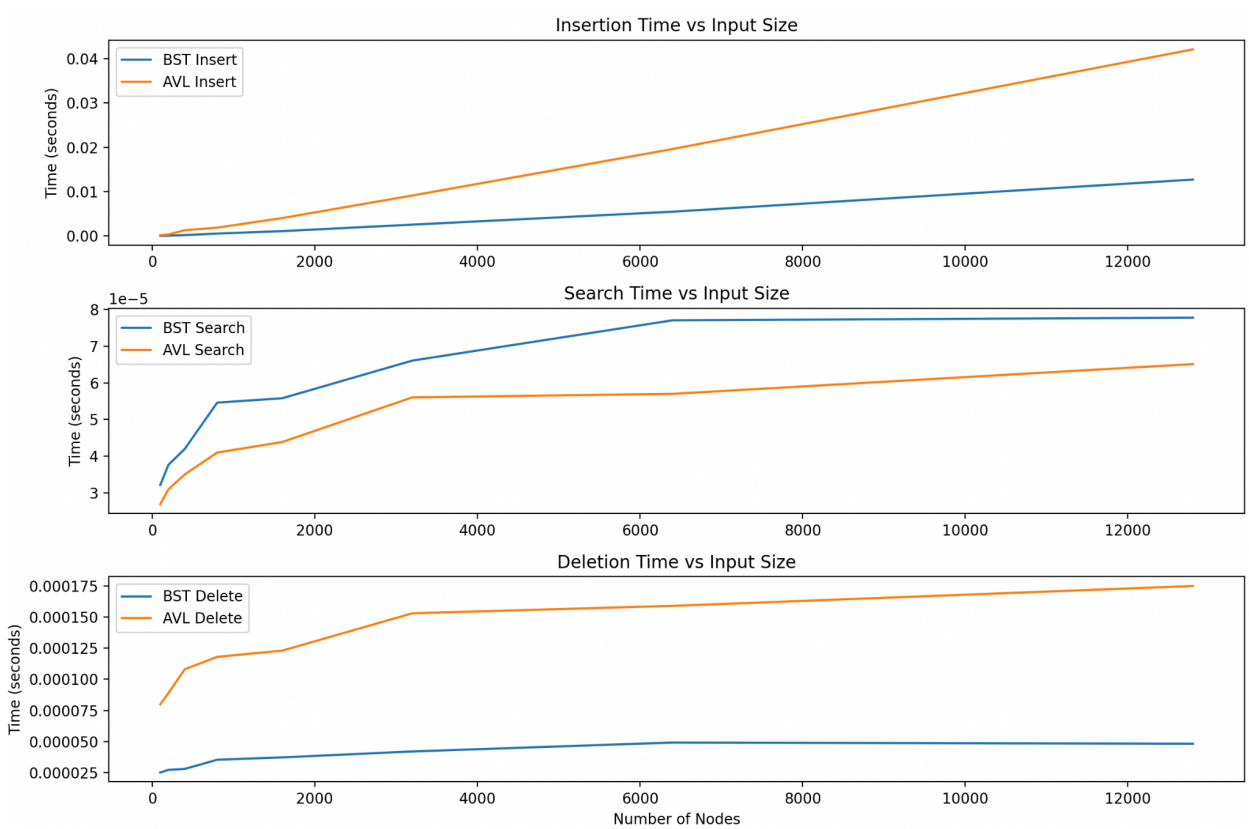
### 5000



10000




Visualiser (extra values 3200, 6400, 12800)



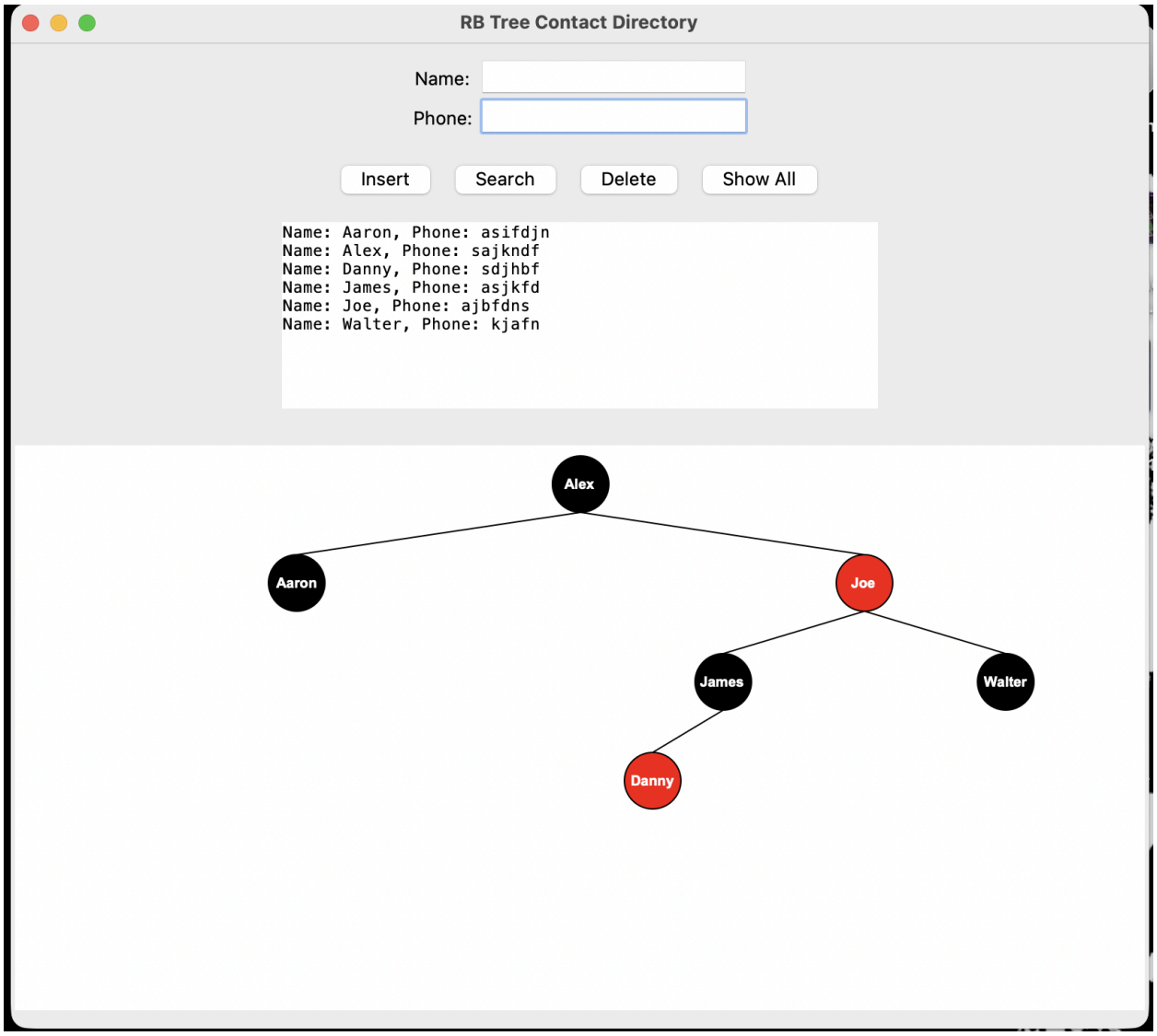

The graphs do support facts about BSTs and AVL trees. Insertion is quicker for the BST, as there is no need for the tree to be sorted once the node is inserted, unlike for AVL trees, which

is reflected in the faster increase of insertion time for AVL trees. AVL trees always have the minimum number of levels possible, which is not a requirement for BSTs, hence the consistently larger time for the search function of BST. Deletion is also much simpler for BSTs, where lack of level requirements mean that fewer child nodes have to be moved, whereas AVL trees have to rotate in order to maintain the correct levels. This is reflected in the much larger times for AVL deletion compared with BST deletion. Overall, these graphs do validate facts about these two types of trees.

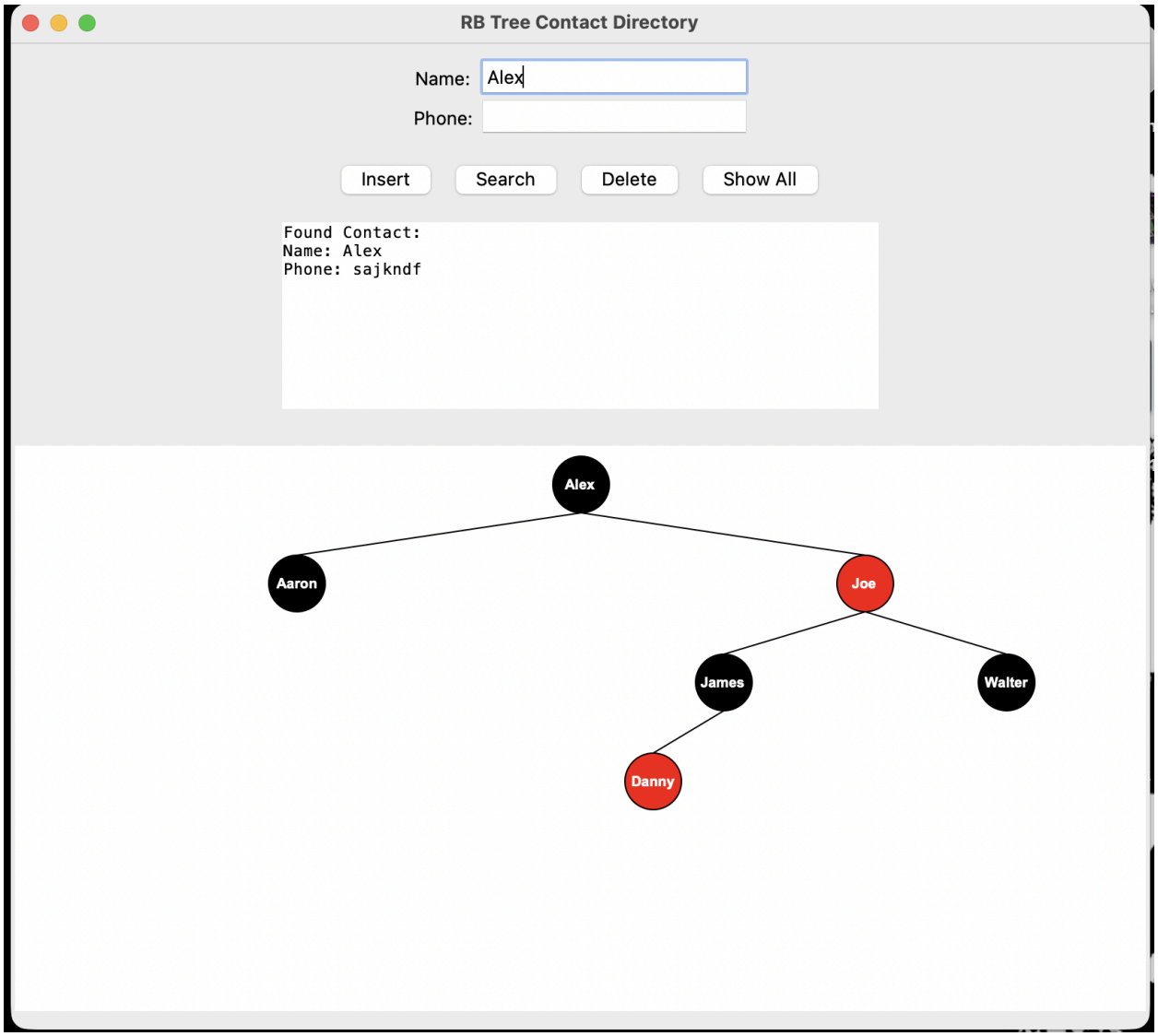
## Task 5:

### Diagram Visualiser

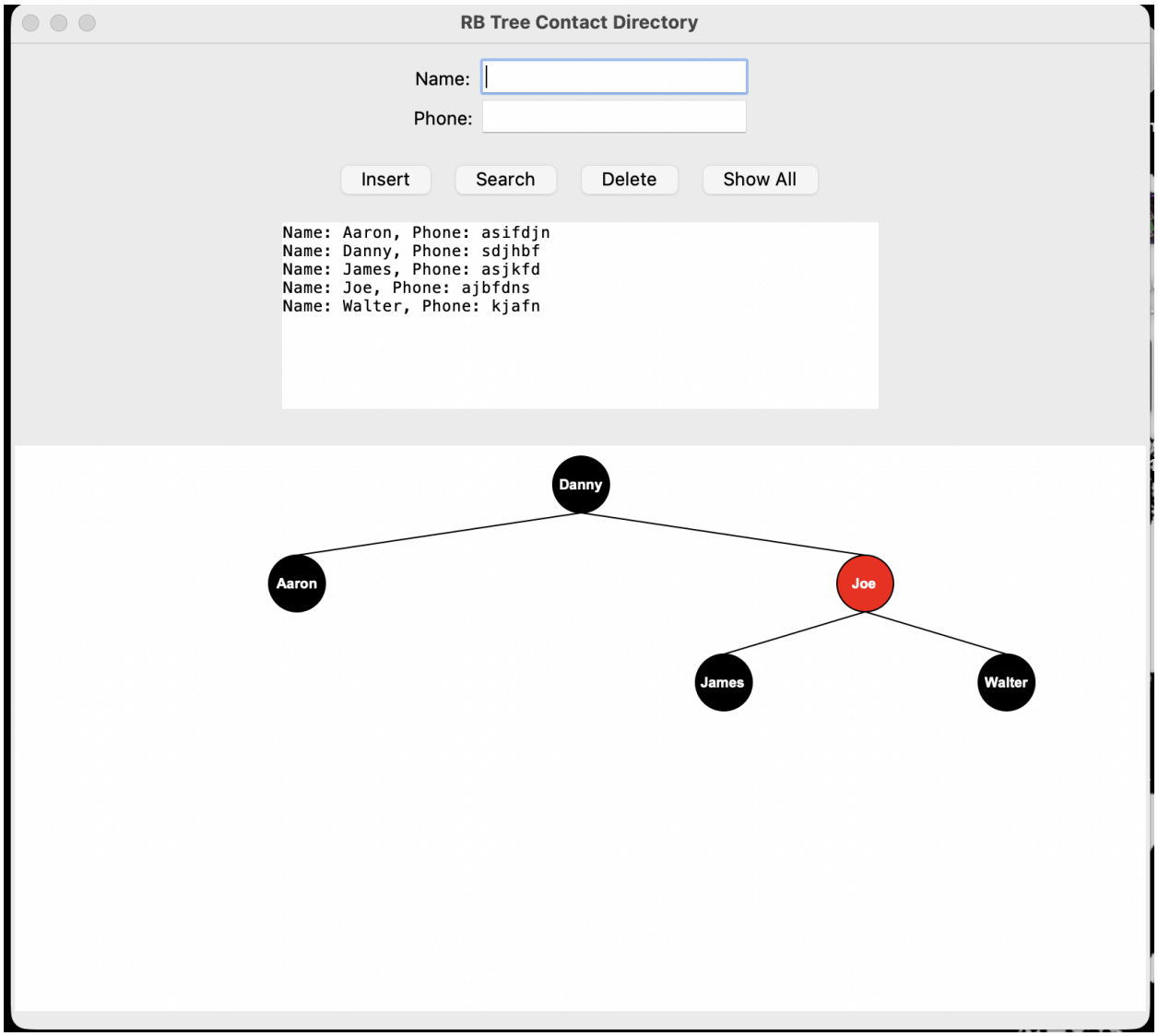
#### Insert



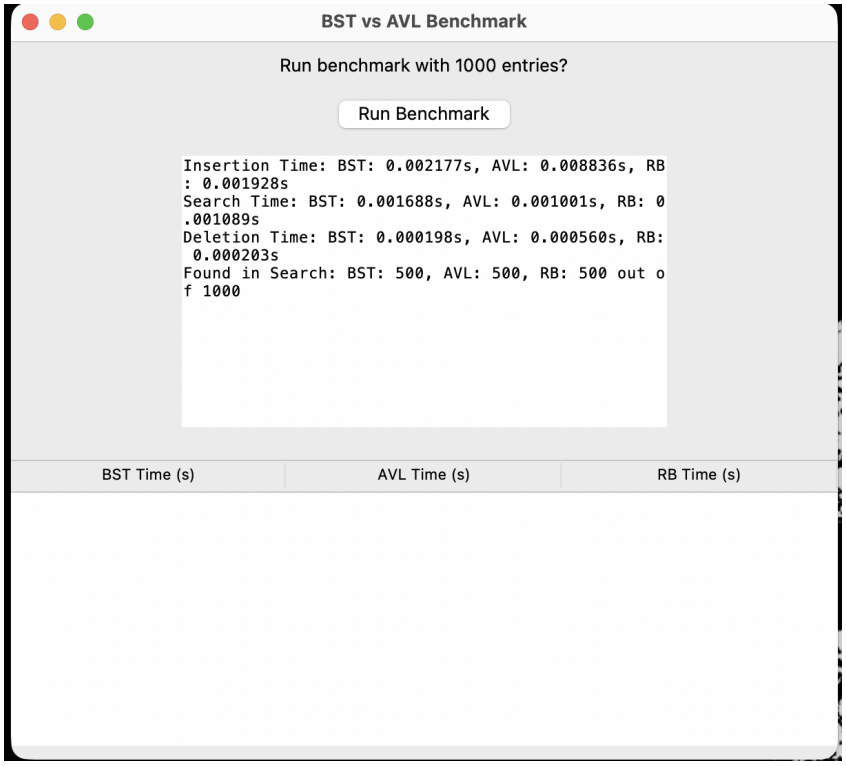
Search

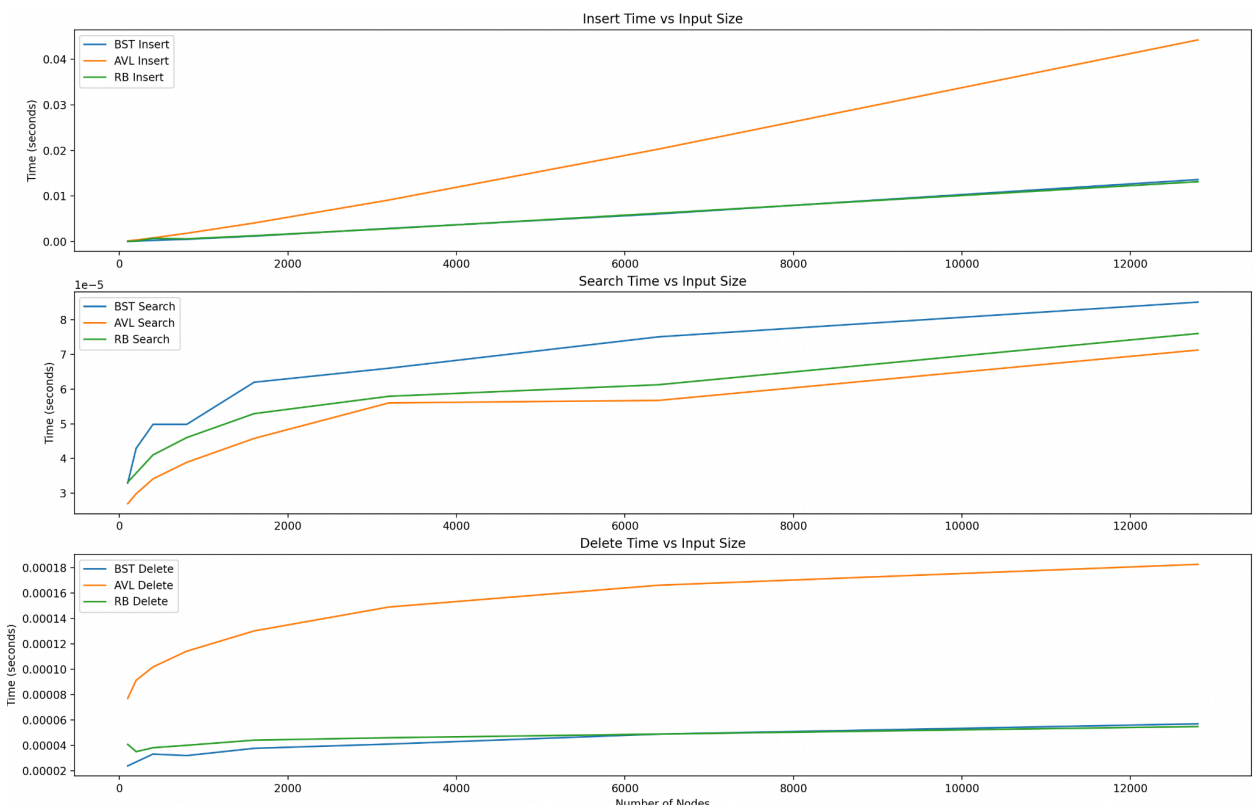

RB Tree Contact Directory
Name:
Alex
Phone:
Insert
Search
Delete
Show All
Found Contact:
Name: Alex
Phone: sajkndf
Alex
Aaron
Joe
James
Walter
Danny

Delete


RB Tree Contact Directory
Name:
Phone:
Insert
Search
Delete
Show All
Name: Aaron, Phone: asifdjn
Name: Danny, Phone: sdjhbf
Name: James, Phone: asjkfd
Name: Joe, Phone: ajbfdns
Name: Walter, Phone: kjafn
Danny
Aaron
Joe
James
Walter

## Benchmarking




## Graph Visualiser